

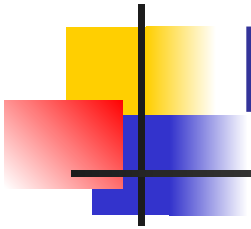
CSE 320
Data Communications

Digital Transmission

Chapter 4

Recap Previous Lecture

- Digital data
  - Digital Signal
  - Analog Signal
- Analog data
  - Digital Signal
  - Analog Signal



Encoding Techniques

- Digital Data, Digital Signal
  - Less expensive and less complex than digital to analog modulation.
- Digital Data, Analog Signals
  - Some transmission media, such as optical fiber and unguided media, will only propagate analog signals.

Digital Data ☐ Digital Signal

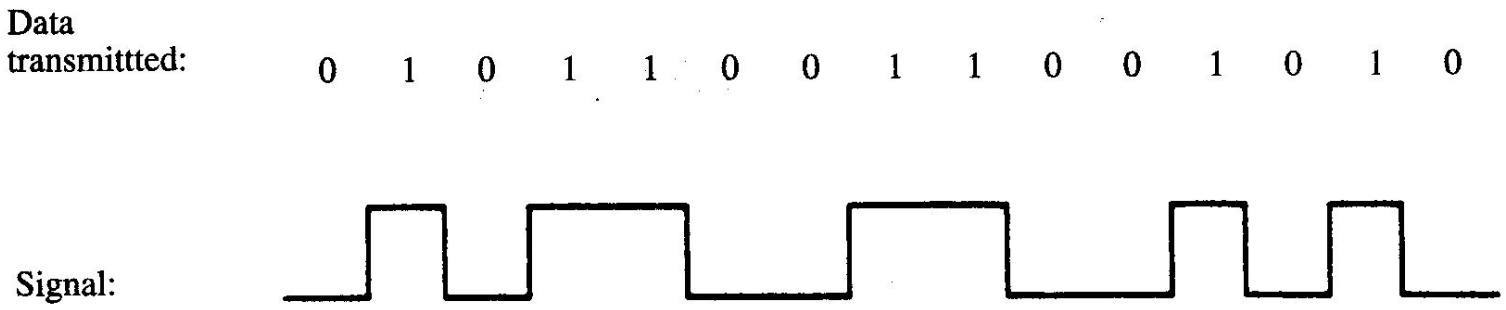
Signal Encoding Techniques:

- **Line Coding**
  - *Diffrent Line Coding Schemes*
- **Block Coding**
- **Scrambling**

Line coding is always needed; block coding and scrambling may or may not be needed.

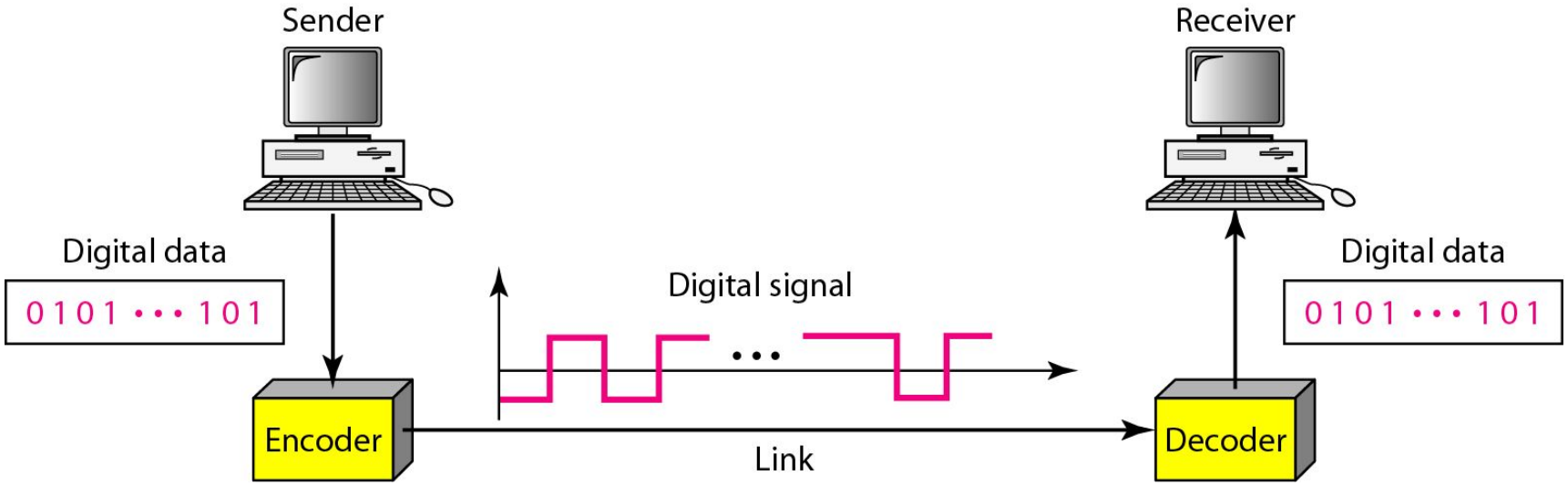
Digital Data, Digital Signal

- Digital signal
  - Discrete, discontinuous voltage pulses
  - Each pulse is a signal element
  - Binary data encoded into signal elements

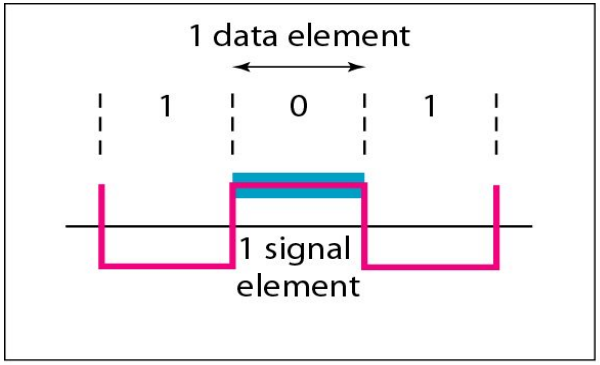


Line Coding

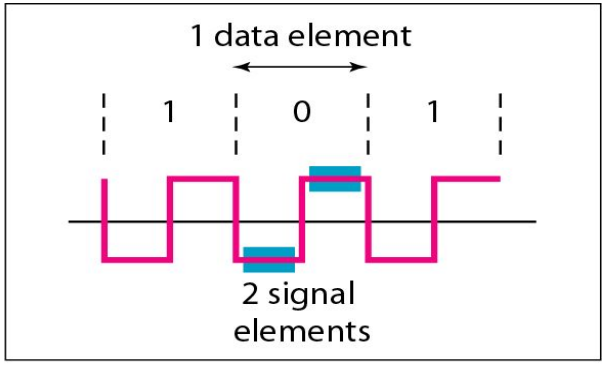
Line coding is the process of converting binary data, a sequence of bits, to a digital signal.



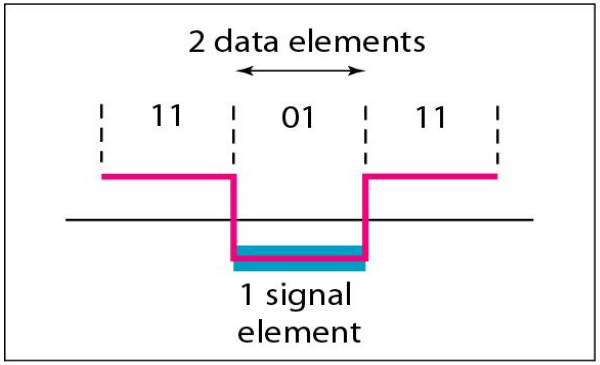
Signal Vs Data Element



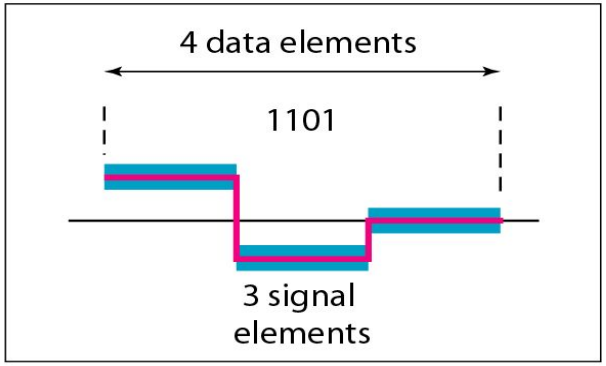
a. One data element per one signal element (r = 1)



b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$



c. Two data elements per one signal element (r = 2)



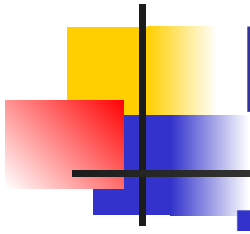
d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

Pulse /Modulation /Signal/ Baud Rate versus Bit Rate

- The pulse rate defines the number of pulses/signals sent in one second. Also known as Baud Rate.
- The bit rate defines the number of bits per second.

$$BitRate = PulseRate \times \log_2 L$$

$$L = Number\ of\ \ data\ levels$$



Digital Data ☐ Digital Signal

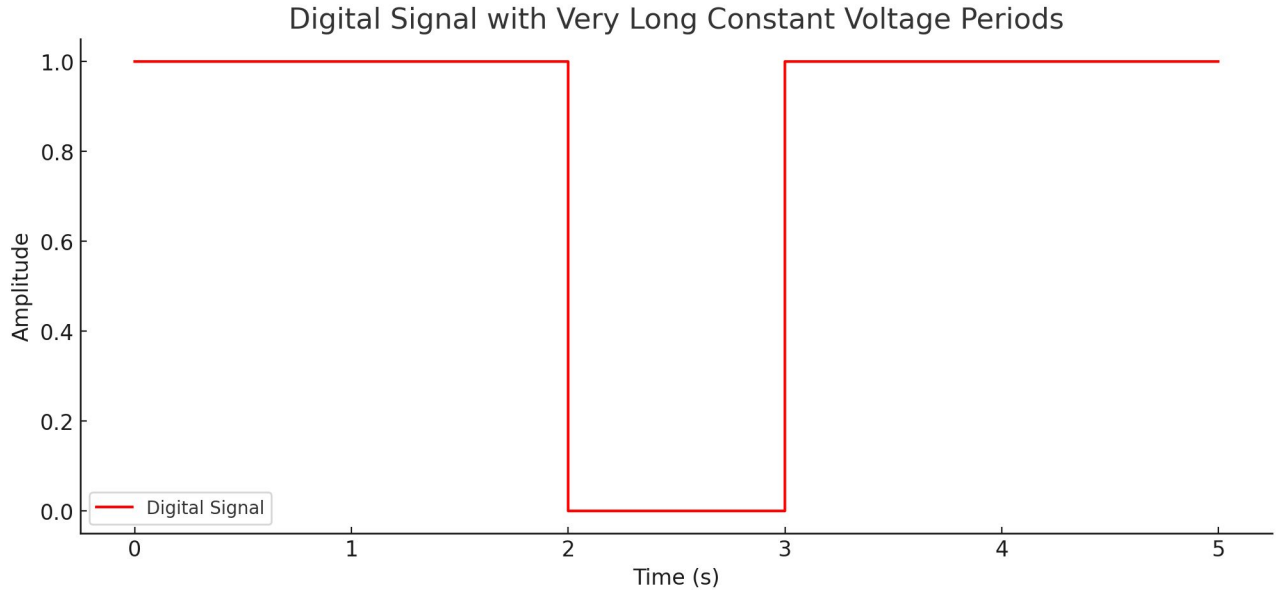
- Receiver needs to know
  - Timing of bits
  - Signal levels
- Factors affecting successful interpretation of signals
  - Baseline Wandering
  - DC Components
  - Self-synchronization
  - Built in Error Detection
  - Immunity to Noise and Interference
  - Complexity

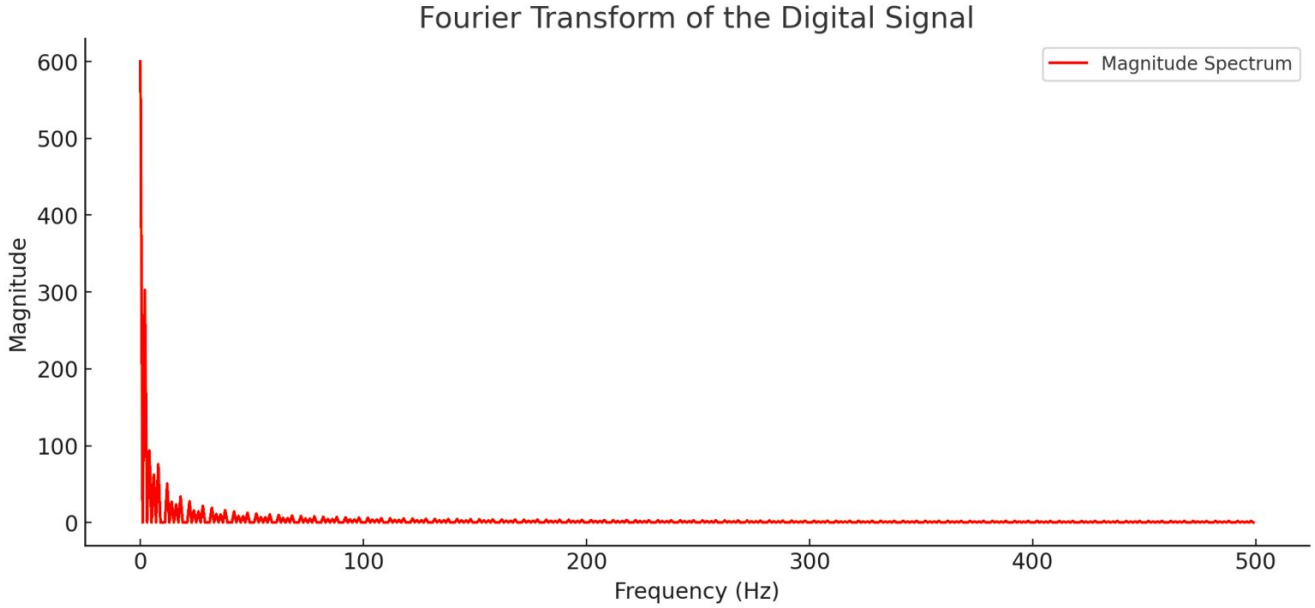
BaseLine Wandering

- Receiver calculates the running average of received signal power.
- Average = baseline
- A log strings of 0's and 1's can cause a drift in the baseline making it difficult for the receiver to decode properly.

DC Components

- When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies (results of Fourier analysis).
- If the signal is to pass through a system (such as a transformer) that does not allow the passage of a dc component, the signal is distorted and may create errors in the output.
- This component is extra energy residing on the line and is useless.

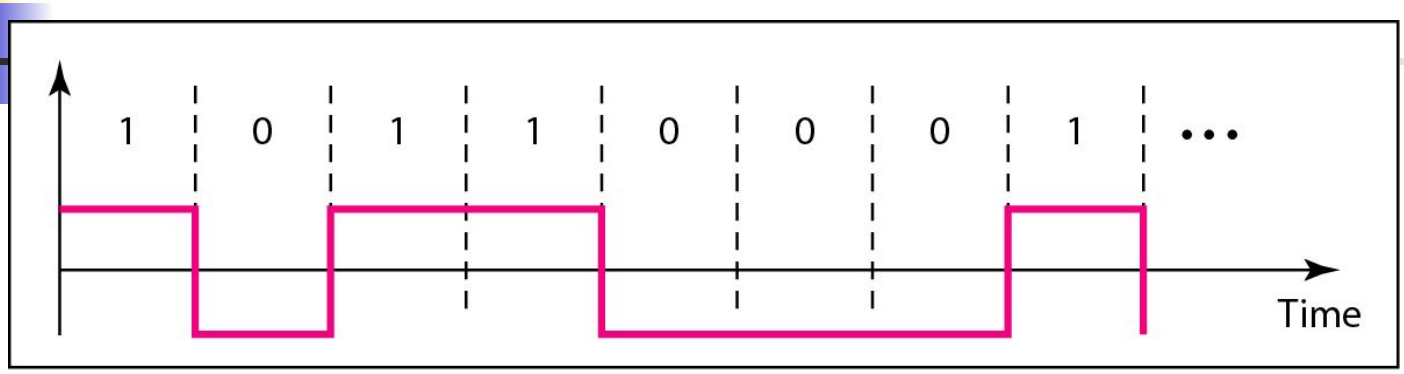

Digital Signal with Very Long Constant Voltage Periods
Amplitude
Time (s)
Digital Signal
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5


Fourier Transform of the Digital Signal
Magnitude
Frequency (Hz)
Magnitude Spectrum
600
500
400
300
200
100
0
0
100
200
300
400
500

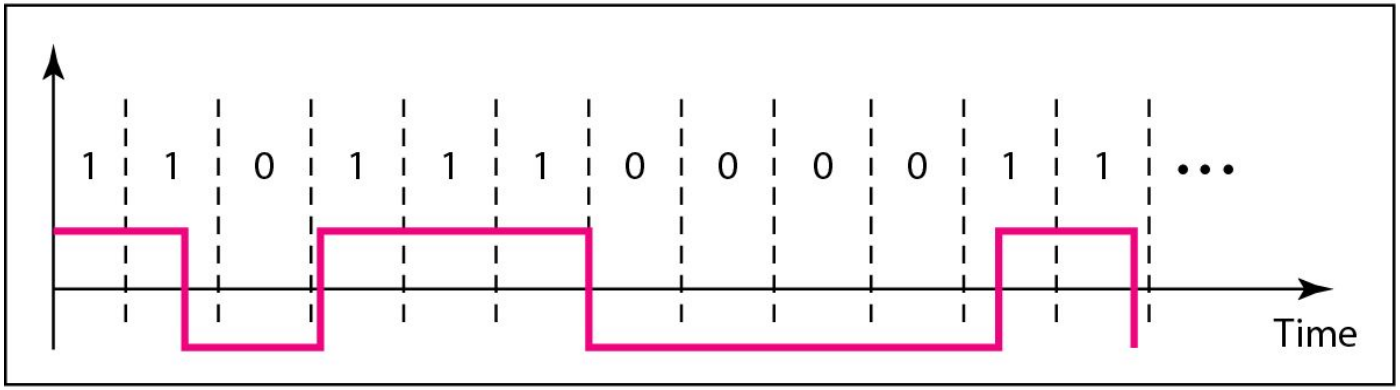
Self-Synchronization

- The receiver's bit intervals must correspond exactly to the senders bit intervals.
- If the receiver clock is faster or slower, the bit intervals are not matched and the receiver might misinterpret the signals.

Figure 4.3 *Effect of lack of synchronization*



a. Sent



b. Received

Self-Synchronization

- A self-synchronizing digital signal includes timing information in the data being transmitted.
- This can be achieved if there are transitions in the signal that alert the receiver to the beginning, middle, or end of the pulse.
- If the receiver's clock is out of synchronization, these points can reset the clock.

Different Line Encoding Schemes

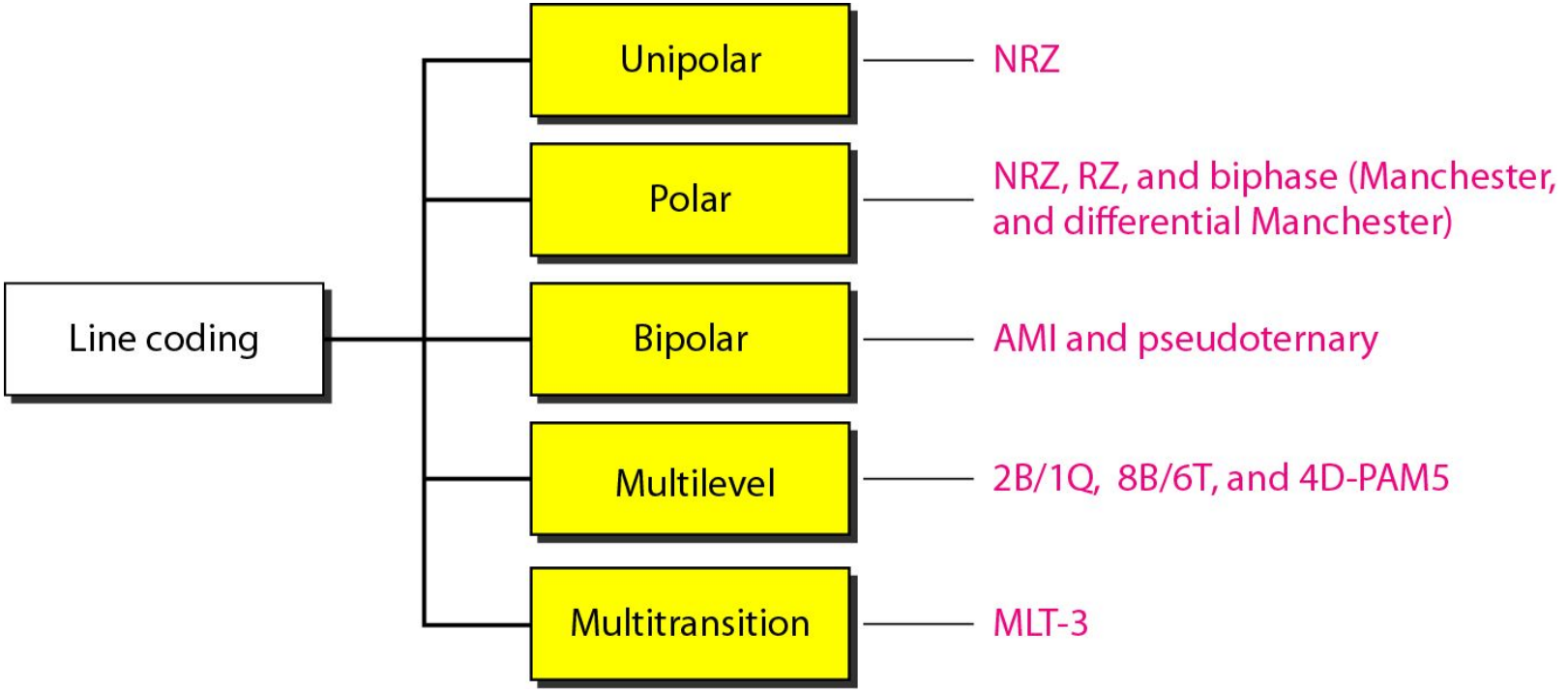
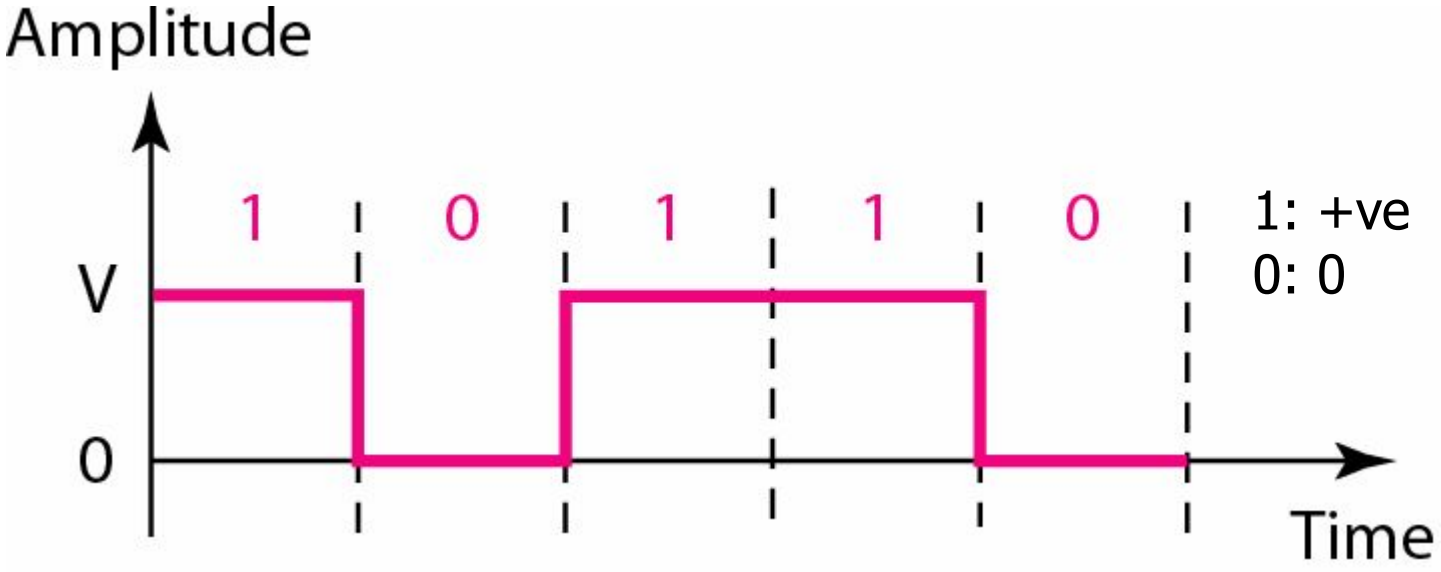


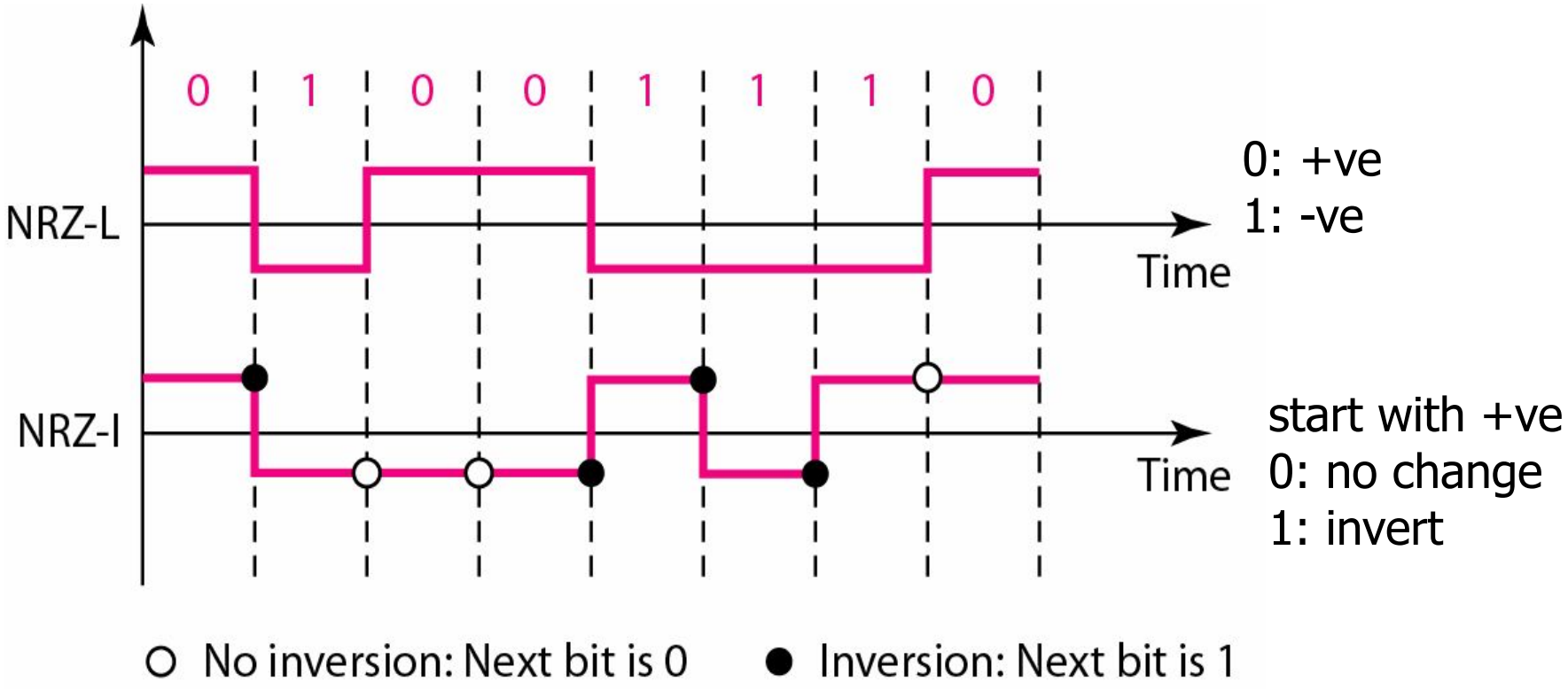
Table 4.1 *Summary of line coding schemes*

Category	*Scheme*	*Bandwidth (average)*	*Characteristics*
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

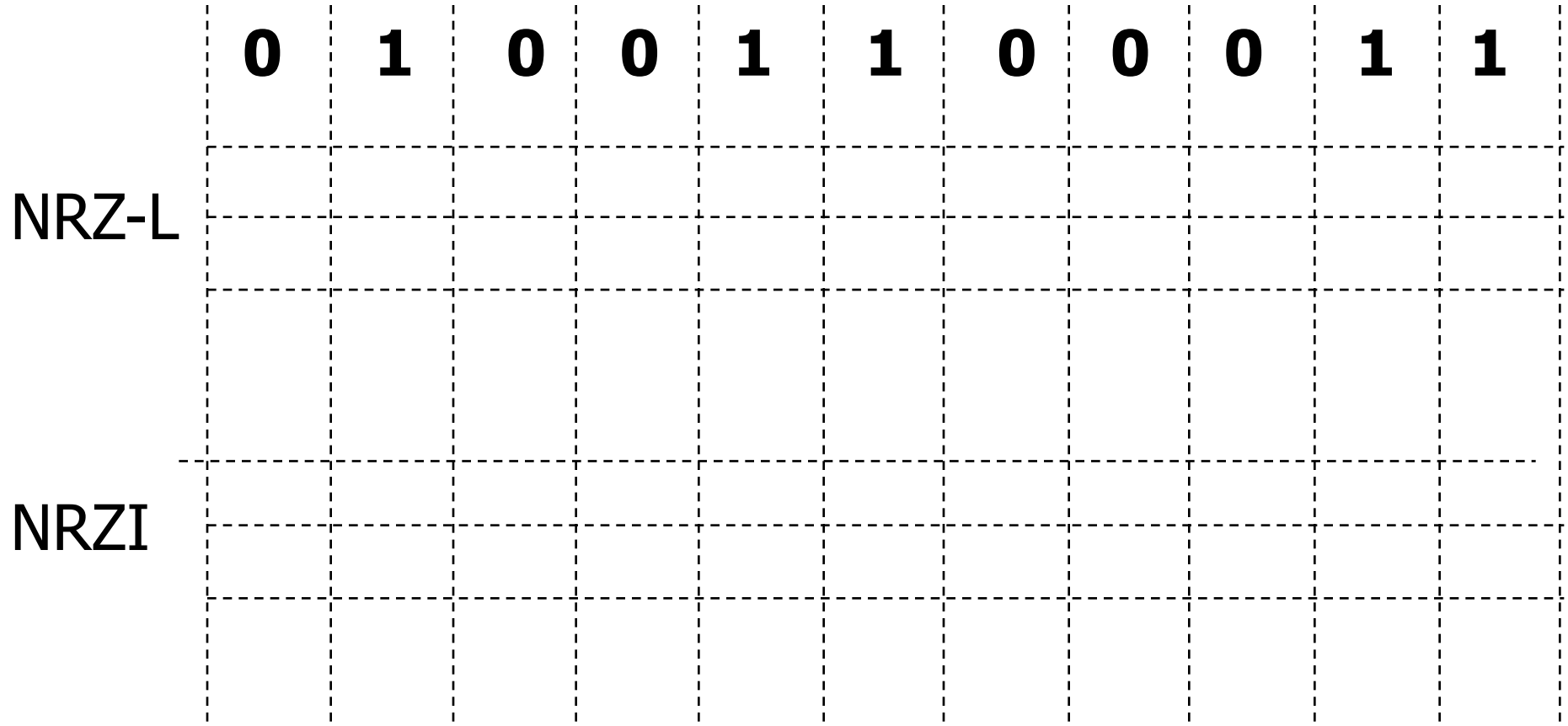
Unipolar



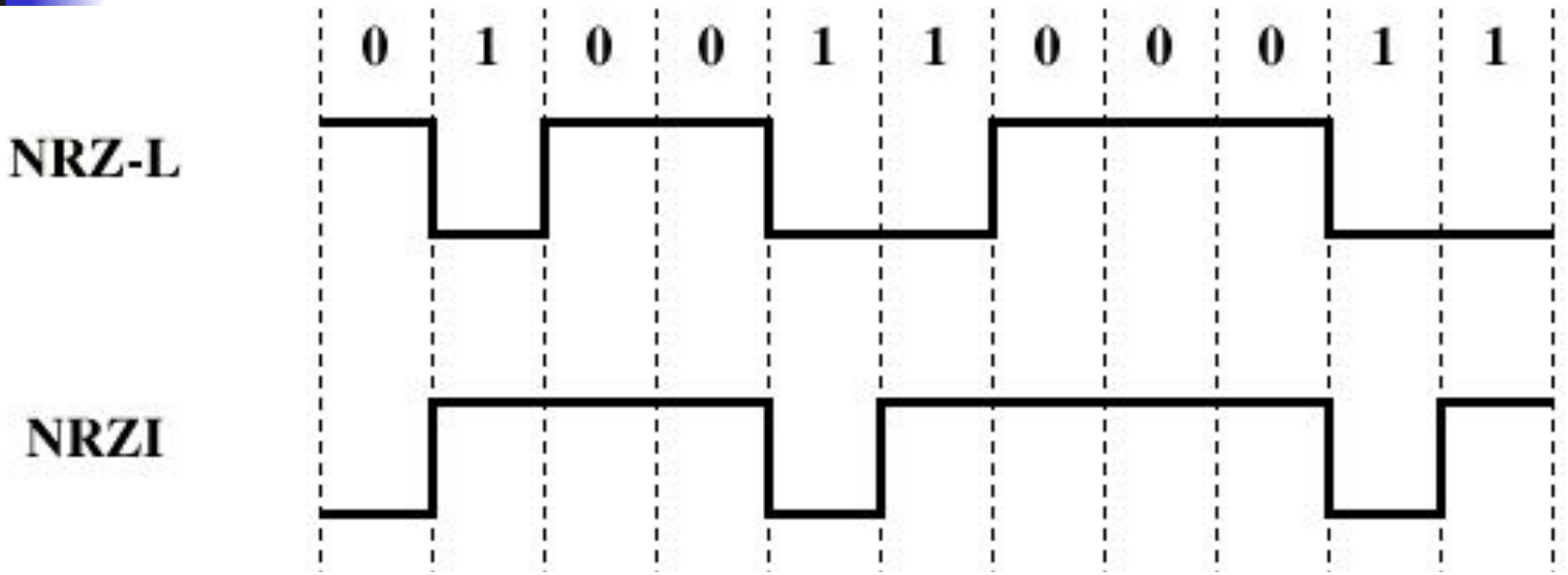
NRZ-L & NRZ-I (Bipolar)



Nonreturn to Zero (NRZ)

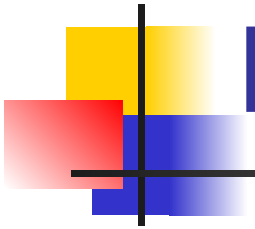


NRZ



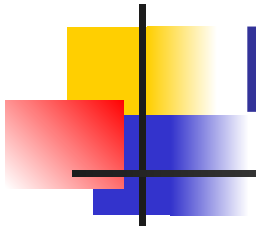
NRZ – Pros and Cons

- Pros
  - Easy to engineer
  - Make good use of bandwidth (r=1)
- Cons
  - DC component
  - Lack of synchronization capability
- Used for magnetic recording
- Not often used for signal transmission



NRZ-L vs NRZ-I

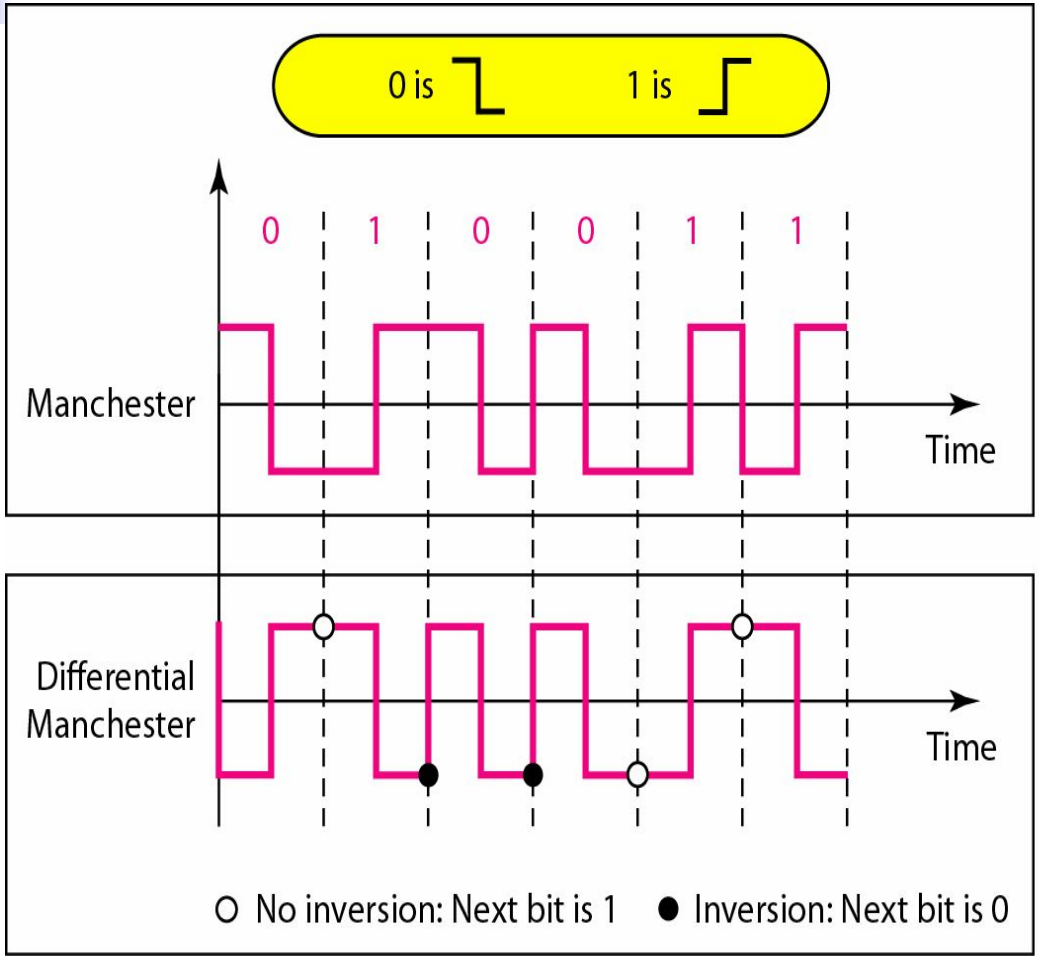
- NRZ-L
  - Baseline wandering, DC components
- NRZ-I
  - No baseline wandering and DC components for consecutive 1s
  - Still exist for consecutive 0s



Differential Encoding

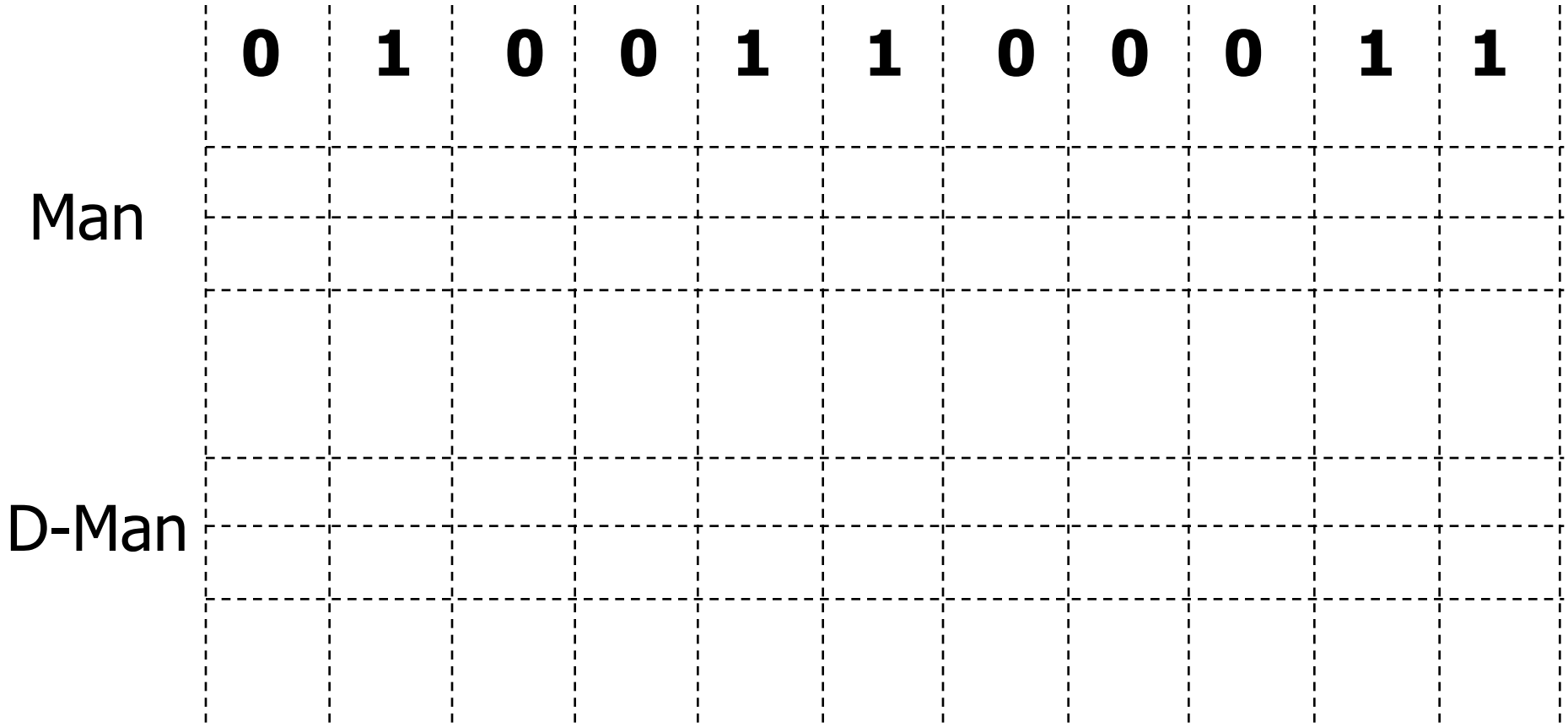
- In complex transmission layouts, it is easy to lose sense of polarity
- Therefore
  - Data represented by changes (i.e., transitions) rather than levels
  - More reliable detection of transition rather than level

Manchester Encoding & Differential Manchester Encoding



start with +ve
0: transition to opposite level
1: no transition

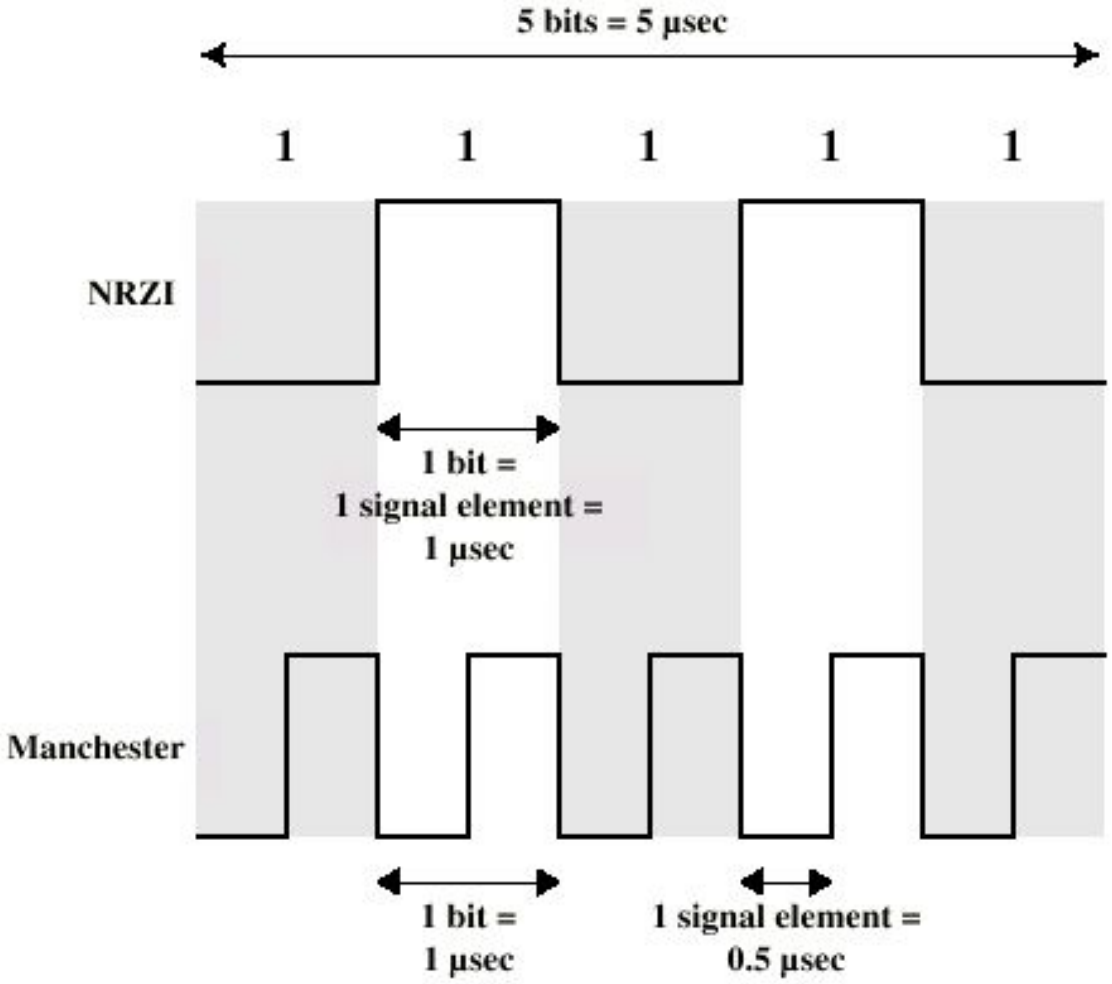
Biphase (Manchester and D-Manchester)

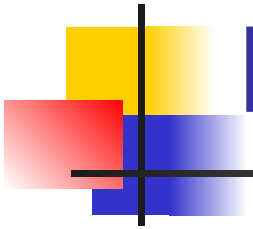


Biphase -- Pros and Cons

- Pros
  - Synchronization on mid bit transition (self clocking)
  - No dc component or baseline wandering
  - Error detection
    - Absence of expected transition
- Cons
  - At least one transition per bit time and possibly two
  - Maximum modulation rate is twice NRZ
  - Requires more bandwidth (r=0.5)

Modulation Rate

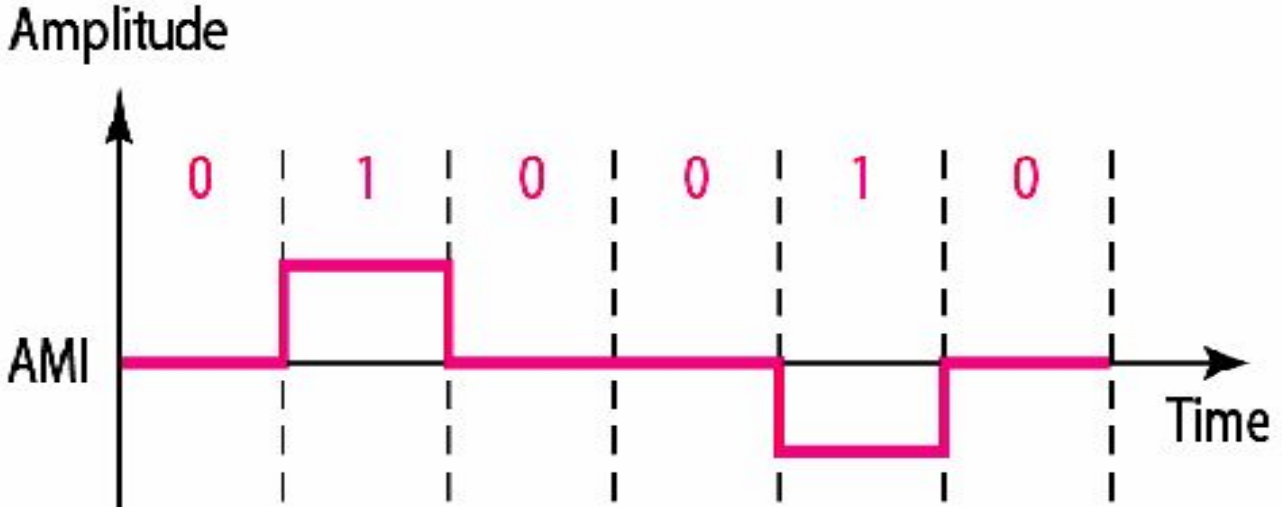




Multilevel Binary

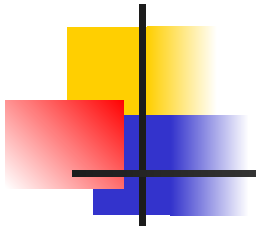
- Use more than two levels
- Bipolar-AMI
  - zero represented by no line signal
  - one represented by positive or negative pulse
  - No baseline wandering for consecutive 1s (zeros still a problem)
  - Good use of bandwidth

Bipolar-AMI



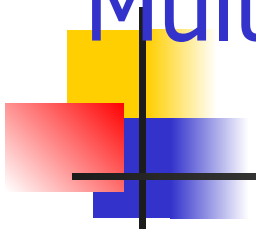
- 0: 0
- 1: opposite of previous non-zero level
- First 1 is +ve

Bipolar AMI



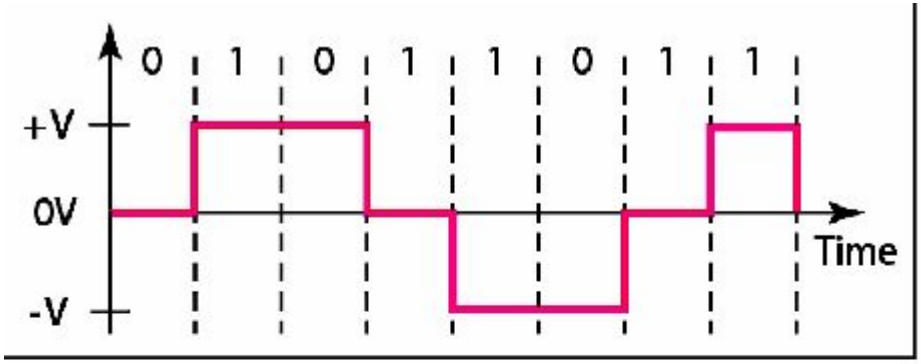
- Baseline wandering for consecutive 0s but not for 1s
- No DC components issue
- Lack of synchronization for a long sequence of 0s
- Good use of bandwidth (r=1)

Multilevel Transition, three level- MLT-3

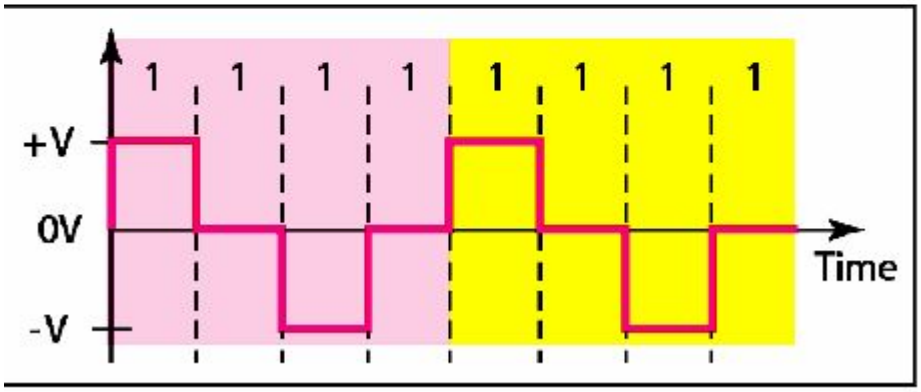


- There is no transition for a 0 bit.
- The signal transitions from one level to the next at the beginning of a 1 bit
- Transition occurs using three levels of signals (+1, 0, -1)
- Start with 0
- First 1 is positive, next 1 is zero, next 1 is negative, next 1 is zero, next 1 is again positive, and so on
- Staircase pattern

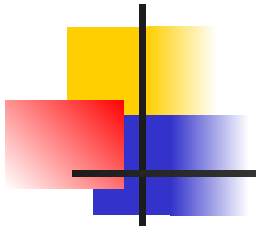
MLT-3



a. Typical case



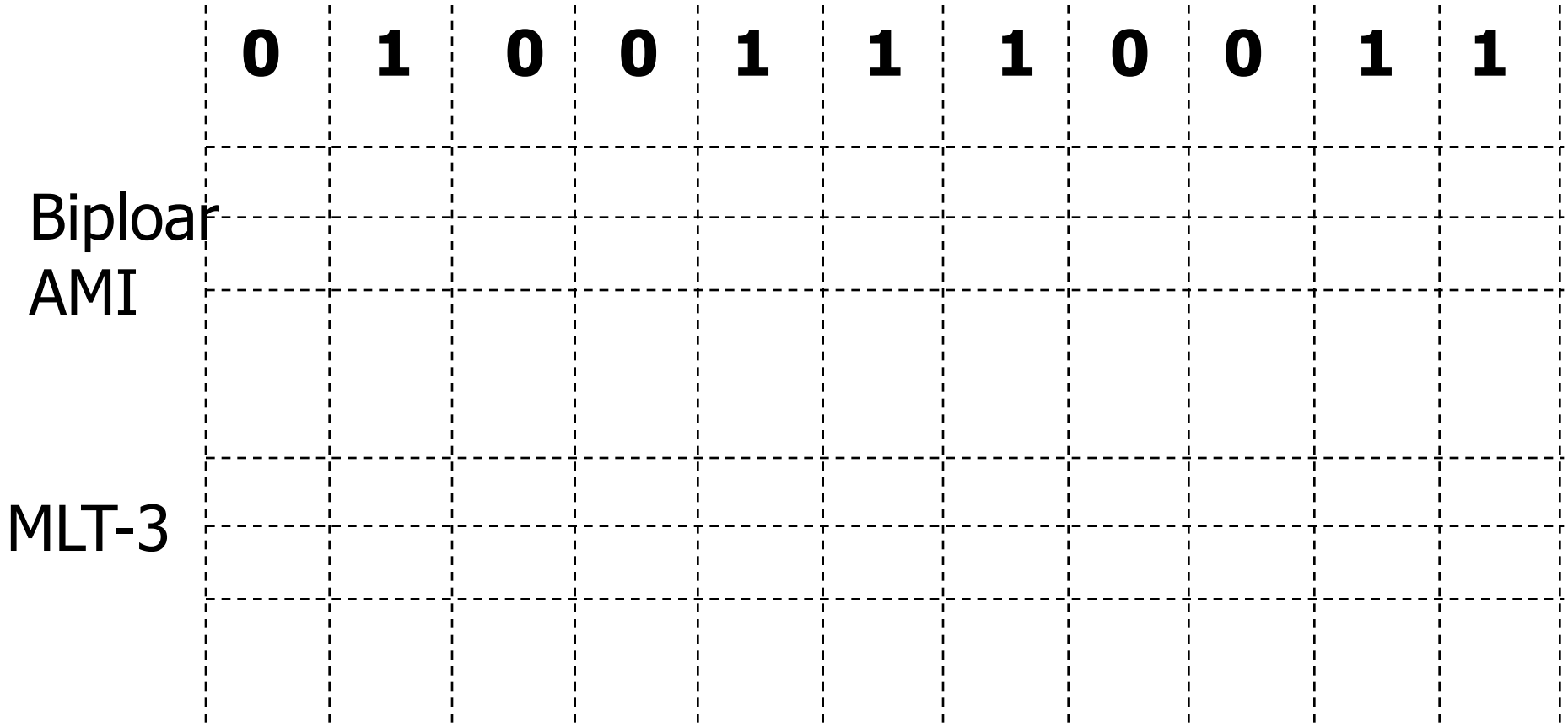
b. Worse case

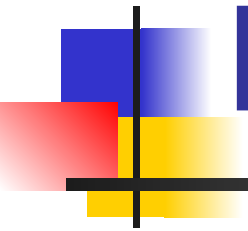


MLT-3

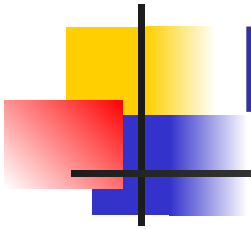
- Baseline wandering for consecutive 0s only, not for 1s
- DC component is an issue
- Lack of synchronization
- Good use of bandwidth (r=1)
  - This is similar to AMI but actually more efficient in practical applications

Biploar AMI and MLT-3 Example



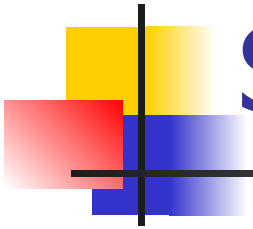


Block Coding



Block Coding/Scrambling

- NRZ, Bipolar AMI, MLT-3 all has a common problem.
- Long sequence of 0 can make the receiver lose synchronization
- Solutions:
  - Change the bit stream before encoding with NRZ-I so that there is no long streams of 0s.

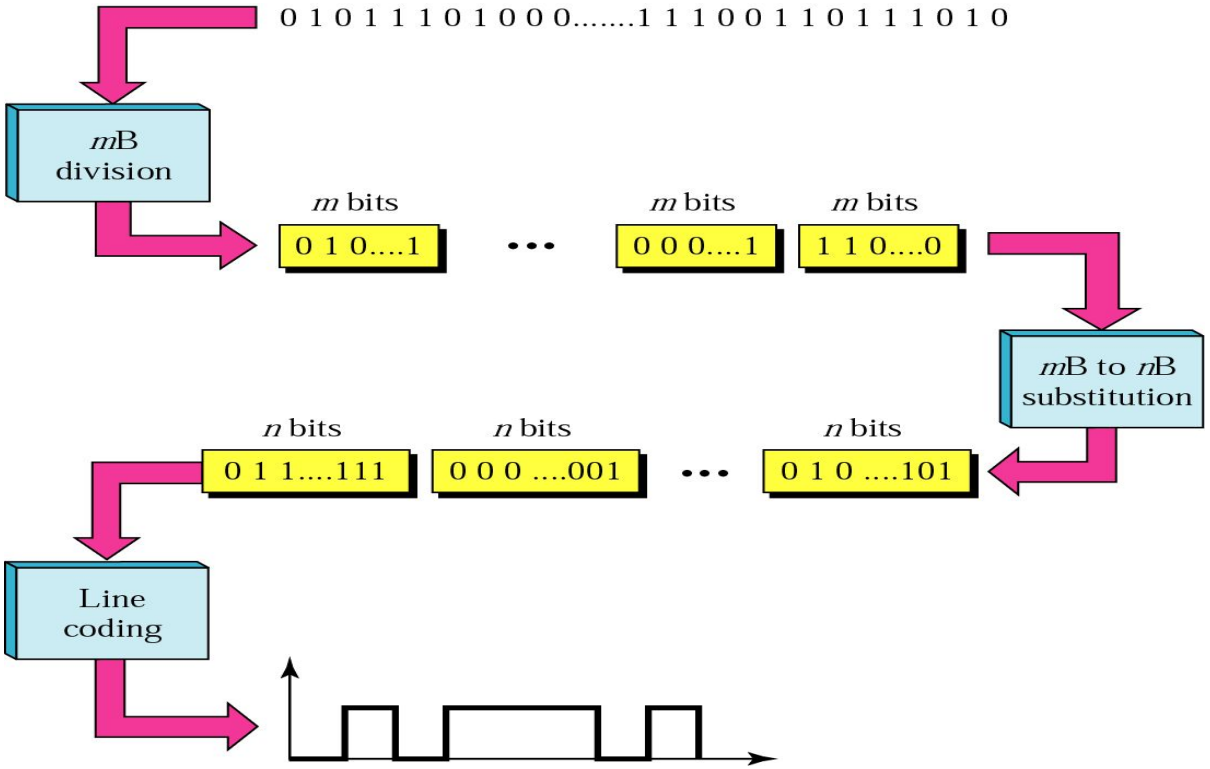


Solutions:

- Block Coding
- Scrambling
- Block Coding
  - Changes a block of m bits to a block of n bits.
  - Referred to as mB/nB encoding.

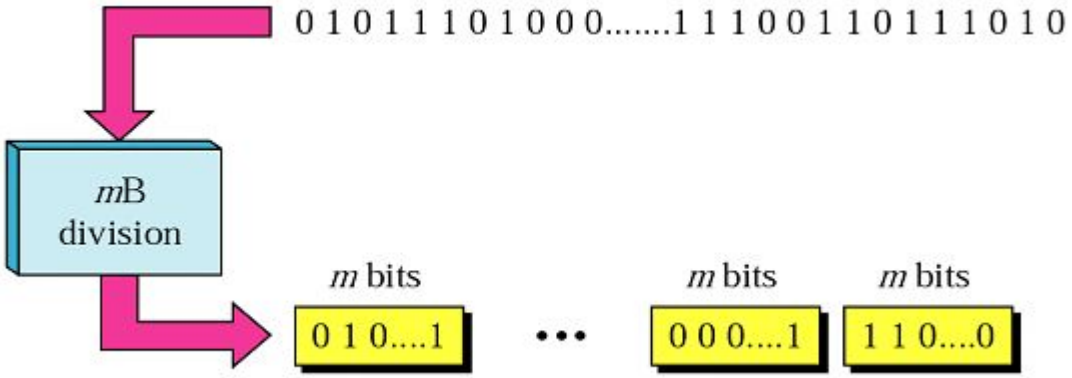
Block Coding

- Three Steps Process:
  - Division
  - Substitution
  - Line Coding /Combination



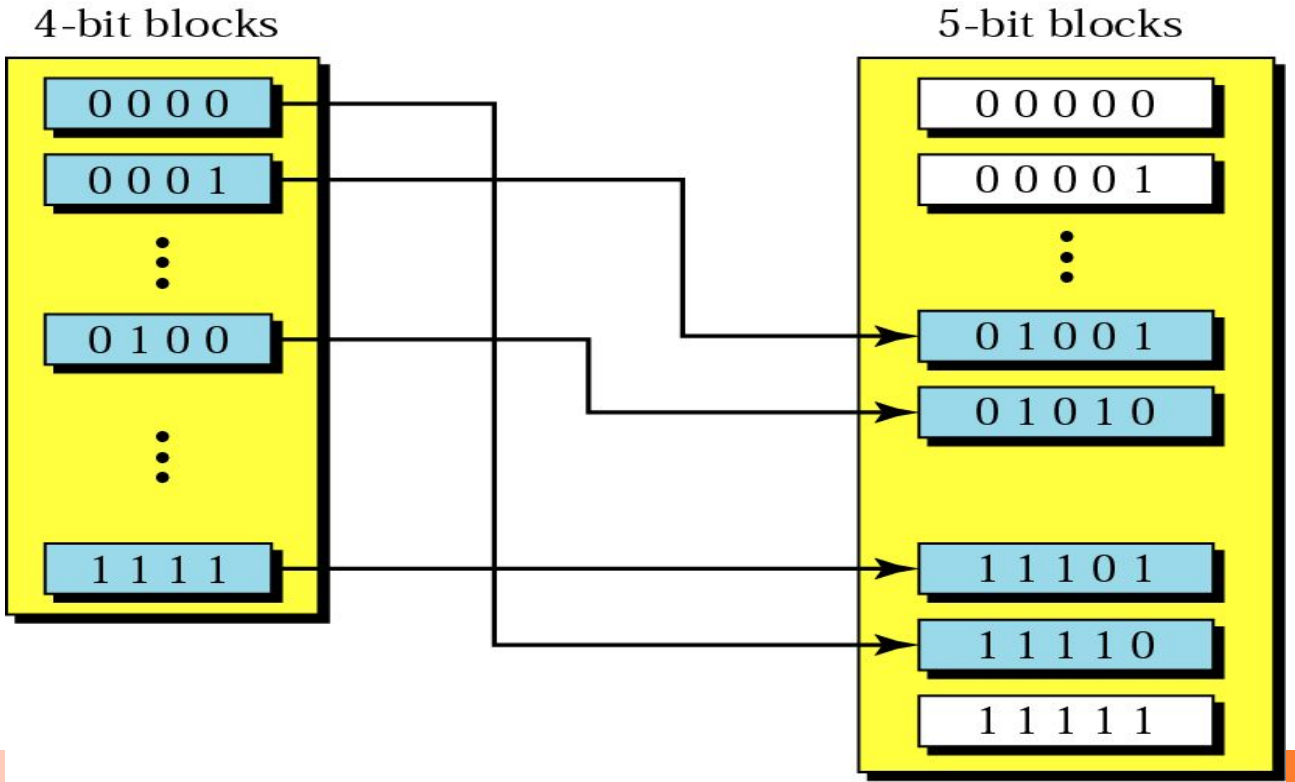
Step 1-Division

- The sequence of bits in data in divided into m Bits.
- For example in 4B/5B encoding, the original bit sequence is divided into 4-bit codes/sequence.



Step 2-Substitution

- Each m bits sequence is substituted for a n bit code.



4B/5B Block Coding

- 4-bit code ->16 different combinations
- 5-bit code -> 32 possible combinations.
- So not all of 5-bit codes are required.
- Selection of the 5-bit code is such that each code contains no more than
  - “one leading 0 and no more than two trailing 0s.” (3 consecutive 0s)

Table : 4B/5B encoding

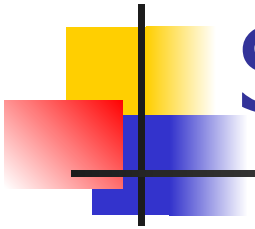
Data Sequence	*Encoded Sequence*	*Control Sequence*	*Encoded Sequence*
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Step 3: Line Coding

- After substitution, a line coding scheme, exp NRZ-I is chosen to create a signal.
- A very simple line coding scheme is chosen, because the block coding procedure provides
  - two desirable features (??) of complex line coding schemes.

Block Coding-Pros/Cons

- Solves the synchronization problem but not the DC component problem.
- If DC is unacceptable, use bipolar or biphase encoding.
- Increases the baud rate by 20%, still better than Manchester schemes.



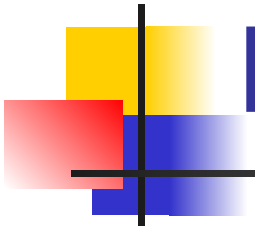
Scrambling

- Use scrambling to replace sequences that would produce constant voltage
- Filling sequence
  - Must be recognized by receiver and replace with original
  - Same length as original
- Design Goals
  - No dc component
  - No long sequences of zero level line signal
  - No reduction in data rate
  - Error detection capability



Types of Scrambling:

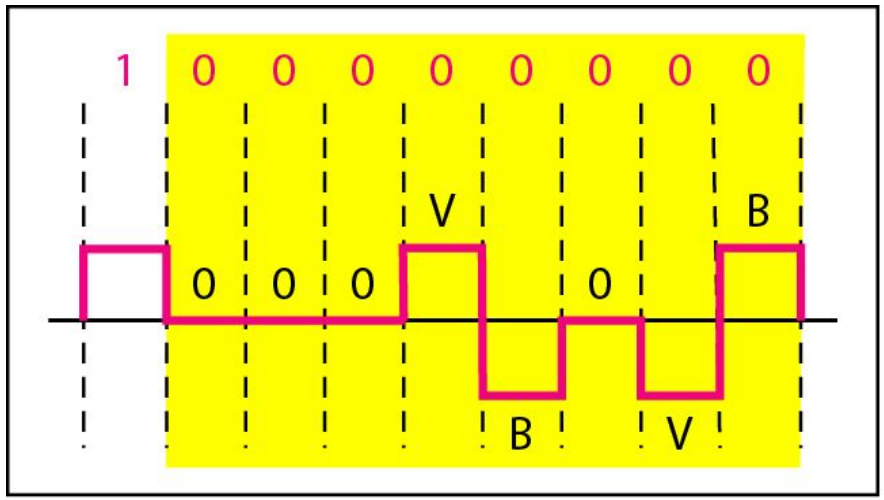
- B8ZS
  - Bipolar With 8 Zeros Substitution
  - Commonly used US.
- HDB3
  - High Density Bipolar 3 Zeros
  - Based on Bipolar AMI
  - Commonly used Europe and Japan.



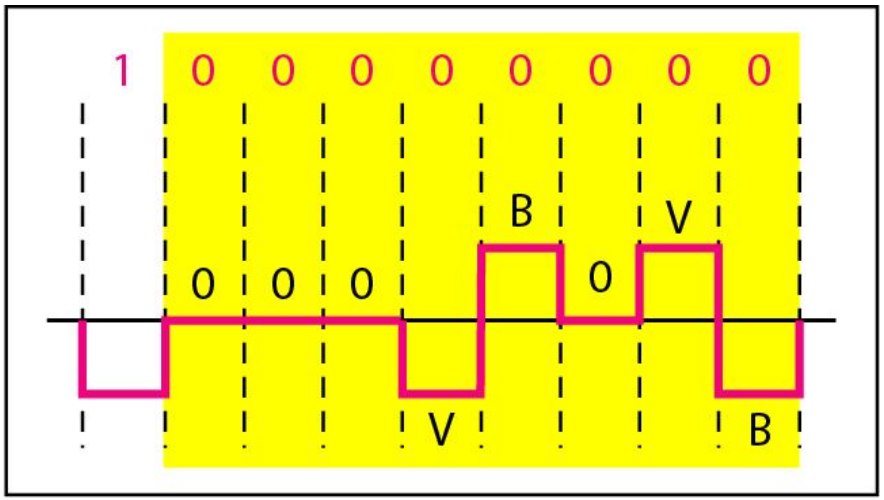
B8ZS

- Based on bipolar-AMI
  - If octet of all zeros and last voltage pulse preceding was positive encode as 000+-0-+
  - If octet of all zeros and last voltage pulse preceding was negative encode as 000-+0+-
- Causes two violations of AMI code
  - Unlikely to occur as a result of noise

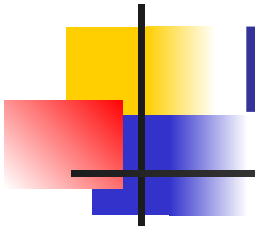
B8ZS



a. Previous level is positive.



b. Previous level is negative.



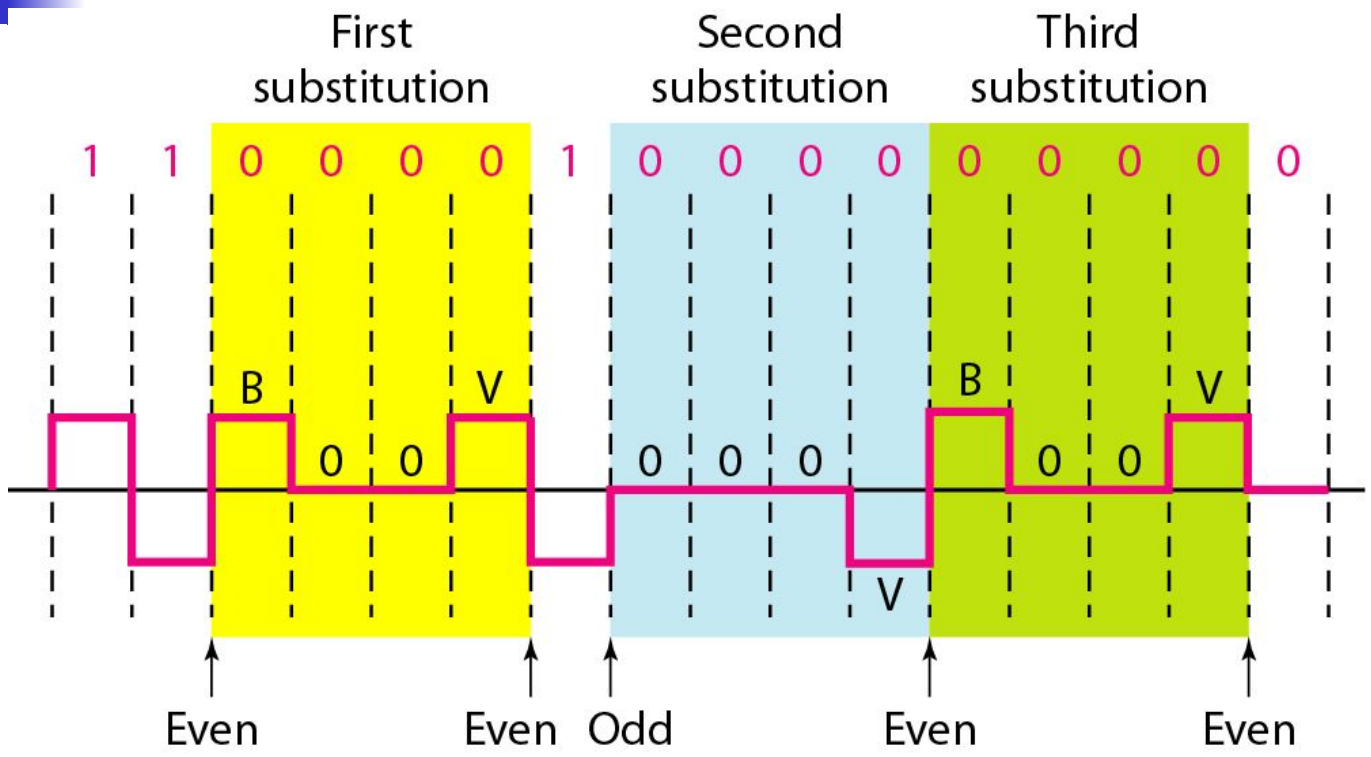
HDB3

- High Density Bipolar 3 Zeros
- Based on Bipolar AMI, Commonly used Europe and Japan.
- Based on bipolar-AMI
- String of four zeros replaced with one or two pulses

HDB3 Subtitution Table

	Number of Bipolar Pulses (ones) since Last Substitution	
Polarity of Preceding Pulse	Odd	Even
–	000–	+00+
+	000+	–00–

HDB3


First substitution
Second substitution
Third substitution
1 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0
B V 0 0 0 0 0 B V 0 0
V
Even Even Odd Even Even